La caja negra: Systems Design

Kevin Vicent · Reto 2 · v4

> **Thesis:** The black box is an unknown decision procedure. We find its rules with experiments. We prove each rule against the real service. Then we compile the rules into a deterministic replica. Most agent systems need an LLM judge. Our system has an exact judge: the original service.

1. Problem

A legacy **ledger service** receives text messages on `POST /msg`. The body is `{"m": "CMD;arg…;control"}`. The commands are `PING`, `ALTA`, `CONSULTA`, `DEPOSITO`, `TRANSFER` and `ANULA`. The service also has `/__reset` and `/health`.

We must build a replacement. After `/__reset`, each response must have **the same status and body** as the original. The comparison ignores JSON key order and **ignores headers**. There are 100 hidden tests: 40 single requests, 45 sequences of 2–5 steps and 15 sequences of 6–12 steps. Every quirk is part of the contract (Hyrum's Law), for example `"recieved"`, `"comando desconocido "` and `cuentaId` vs. `cuenta_id`. The replica must not call the original. **The service changes during the day. The tests use the 20:00 version.**

Assumptions (we test each one): A1 The output depends only on state and request. **A2** Each token has its own state. `/__reset` restores the initial state (`AC-0001`, `OP-0001`). **A3** (*probable*) The control is a **mod-97 family checksum on the arguments only**. Observed values go up to 98, so we expect an IBAN-style transform (`98 − x`). The service checks it **before** the account lookup (a bad control on a missing account gives E202, not E401). **A4** Tests are sequential. **Hard cases:** the control formula; **error precedence** (E2xx integrity, E301/E302 syntax, E401 lookup, E506/E507 business rules); ID sequences; `ANULA` twice; limits; unknown routes; **drift**. **Out of scope:** headers and latency (not tested).

2. Architecture: dual-process, one ontology

1. **Agent harness (System 2, offline).** A person and a reasoning model use only recorded tools: `probe`, `sequence`, `fuzz`, `shrink`, `diff`. Each request becomes one line of `experimentos.jsonl`, with its hypothesis and X–Req. **Rule: no request without a prediction.** A code gate (a `beforeToolCall` hook) blocks a request if it has no predicted response, if the log already predicts it, or if it exceeds the budget.
2. **Hypothesis search, not answer search (active learning).** Our first sweep tried controls 0–99 and used 543 requests. Now the harness keeps a set of candidate rules. It fits them offline on the log, at zero cost. Then it sends the one request that **splits the surviving candidates in half**. This needs about $\log_2(H)$ requests, approximately 10–15.
3. **Service ontology (memory).** Each rule is typed: `Rule{command, when:[ordered preconditions], then:{status, body}, effect, evidence:[log ids], confianza}`. The checksum is a typed slot: `Checksum{scope, mapping, base, modulus, transform}`. The same table does three jobs. The replica **executes** it. `SPEC.md` **is generated** from it (7 sections, `confirmado / probable / supuesto`). The tests come from its evidence.
4. **Replica (System 1, runtime).** HTTP adapter → serial queue → decision core `handle(state[token], msg) → (state, resp)`. Order: route → auth → parse → control → command → arity → lookup → business rule → effect. State is in memory.

Decision	Chosen	Rejected, and why
Runtime	**Compiled rule table**	*LLM in the replica (allowed, ≤ 5 s):* not exact, not repeatable. *Record and replay:* fails on new inputs.
Discovery	**Person + model in a harness**, exact oracle	*Free LLM chat:* no proof, no log. *Autonomous agent:* false spec claims **lose points**. *Manual curl:* too slow.
Knowledge	**Typed ontology**, one source for code, spec and tests	*Separate spec, code and tests:* they diverge after the change. *RAG over logs:* approximate recall.
Checks	**Code for all exact checks** (budget gate, checksum, rule replay). A **System One classifier** only for drift triage and model routing. It fails closed.	*LLM judge as verifier:* false passes. The original is a free, exact judge.
Server and hosting	**Minimal server we control** (exact bodies, also for unknown routes); **one always-on instance**	*Framework defaults:* their 404/405 bodies are different. *Autoscaling:* each instance has its own state.

3. How we measure: evals first

We optimize against evals and we protect them from Goodhart's Law: **corpus** = train, **held-out fuzz** = test, **canary** = "is the eval still true?", **hidden set** = final test.

- **M1 Exact match** with the official comparator. Target: 100% on the corpus and the 20 `ejemplos`; **≥ 95% on held-out fuzz**. We never fix code against held-out fuzz.
- **M2 Coverage:** command × precondition × error matrix. The fuzzer tests **low-confidence rules and empty cells first**. **Budget per token: 8,000 requests/day, 10/s.** Plan: 2,000 discovery, 2,500 fuzz, 500 canary, 3,000 reserve. A teammate's token adds 8,000.
- **M3 Drift:** a **10-request canary** every 10 min (~500/day). A full replay runs only if the canary fails. Detect in ≤ 10 min. Fix in ≤ 45 min.
- **Checksum eval:** the formula must predict all logged controls (train) and new strings with digits, lowercase and accents (test).
- **Seeded tests:** each error alone and **in pairs** (precedence), control ±1, case variants, arity ±1, `ANULA` twice, transfers near the limit.
- **3 validations:** only if M1 did not decrease (~13:30, ~17:30, ~19:30).

Risks: the formula is not in our family (add candidates, still split in half) · budget runs out (cache, never repeat) · late drift (canary runs until 20:00).